

AN EXAMPLE OF NETWORK EQUILIBRIUM FORMULATIONS

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Warning: of direct interest for the course refer only to sections 1, 2, 3, 4, 5.3, 5.4, 6, and 7.

It is out of the scope of the course to give a mathematical proof that problems formulated in 5.1, 5.2, 5.3, 5.4, 5.5, and 5.6 are equivalent each other and that their solutions verify the Equilibrium formulation in 4.

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1 Data of the network used as example

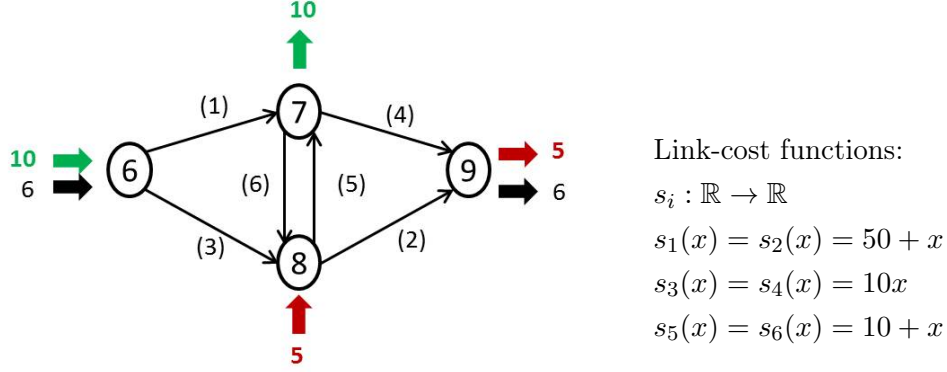
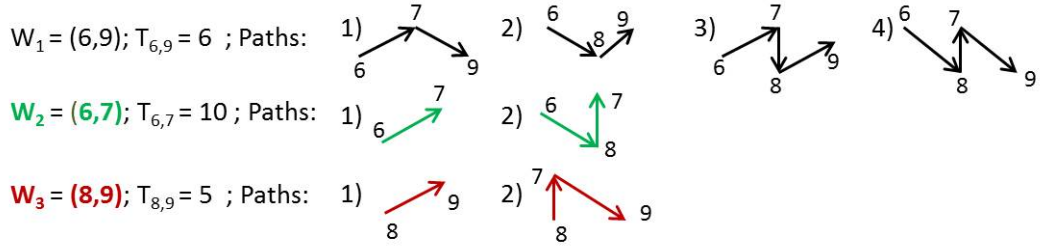


Figure 1: Network definition

$W = \{(6, 9), (6, 7), (8, 9)\}$, set of O-D pairs. $\Gamma_{(6,9)}$ = set of paths joining 6 to 9; $\Gamma_{(6,7)}$ = set of paths joining 6 to 7; $\Gamma_{(8,9)}$ = set of paths joining 8 to 9.

In general, if W is the set of O-D pairs in the network and $w = (p, q) \in W$, by Γ_w it will be denoted the set of all the paths joining p to q .



Notice: only acyclic paths count.

2 Feasible flows: path-flow formulation

Let h be the vector of path-flows, $h = (\dots, h_r^{w_j}, \dots)$, where $h_r^{w_j}$ is the flow of paths r of the O-D pair $w_j, j = 1, 2, 3$. In our example,

$$h^T = (h_1^{w_1}, h_2^{w_1}, h_3^{w_1}, h_4^{w_1}, h_1^{w_2}, h_2^{w_2}, h_1^{w_3}, h_2^{w_3})$$

Let Ω be the feasible set of path flows:

$$\Omega = \{h \in \mathbb{R}_+^8 \mid \underline{h_1^{w_1} + h_2^{w_1} + h_3^{w_1} + h_4^{w_1} = 6, h_1^{w_2} + h_2^{w_2} = 10, h_1^{w_3} + h_2^{w_3} = 5}\}$$

3 Feasible flows: multicommodity link-flow formulation

Each commodity corresponds to an O-D pair.

v_ℓ^k = flow on link ℓ for the O-D pair $w_k, k = 1, 2, 3$

v_ℓ = total flow on link ℓ

Relationship with path flows:

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} h_1^{w_1} \\ h_2^{w_1} \\ h_3^{w_1} \\ h_4^{w_1} \\ h_1^{w_2} \\ h_2^{w_2} \\ h_1^{w_3} \\ h_2^{w_3} \end{bmatrix} \Rightarrow v = \Delta \cdot h$$

3.1 Formulation in commodities per O-D pair

A node-link incidence matrix:

$$A = \left(\begin{array}{c|cccccc} & \overbrace{1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6}^{\text{arcs}} \\ \hline \begin{array}{c} 1 \\ -1 \\ 0 \\ 0 \end{array} & \begin{pmatrix} 1 & 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 & -1 & 1 \\ 0 & 1 & -1 & 0 & 1 & -1 \\ 0 & -1 & 0 & -1 & 0 & 0 \end{pmatrix} & \begin{array}{c} 6 \\ 7 \\ 8 \\ 9 \end{array} \end{array} \right) \begin{array}{c} \text{nodes} \end{array}$$

- O-D pair $w_1 = (6, 9)$:

$$A \cdot \begin{bmatrix} v_1^1 \\ v_2^1 \\ v_3^1 \\ v_4^1 \\ v_5^1 \\ v_6^1 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ 0 \\ -6 \end{bmatrix}, v_i^1 \geq 0, i = 1, \dots, 6$$

- O-D pair $w_2 = (6, 7)$:

$$A \cdot \begin{bmatrix} v_1^2 \\ v_2^2 \\ v_3^2 \\ v_4^2 \\ v_5^2 \\ v_6^2 \end{bmatrix} = \begin{bmatrix} 10 \\ -10 \\ 0 \\ 0 \end{bmatrix}, v_i^2 \geq 0, i = 1, \dots, 6$$

- O-D pair $w_3 = (8, 9)$:

$$A \cdot \begin{bmatrix} v_1^3 \\ v_2^3 \\ v_3^3 \\ v_4^3 \\ v_5^3 \\ v_6^3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \\ -5 \end{bmatrix}, v_i^3 \geq 0, i = 1, \dots, 6$$

3.2 Formulation in commodities per origin

Let v_i^j be flow on link $i = 1, \dots, 6$ originated at j .

- origin 6

$$A \cdot \begin{bmatrix} v_1^6 \\ v_2^6 \\ \dots \\ v_6^6 \end{bmatrix} = \begin{bmatrix} 16 \\ -10 \\ 0 \\ -6 \end{bmatrix}, v_i^6 \geq 6, i = 1, \dots, 6$$

- origin 8

$$A \cdot \begin{bmatrix} v_1^8 \\ v_2^8 \\ \dots \\ v_6^8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \\ -5 \end{bmatrix}, v_i^8 \geq 6, i = 1, \dots, 6$$

Warning: These formulations using matrix A provide solutions containing cycles. Fortunately, solutions with cycles will never appear in the solution of well posed equilibrium problems.

4 Path-based costs for the example network

Each path has an associated path-cost function C . Usually path-cost functions are additive, i.e. are computed as the sum of costs of link-cost functions for all the links making up the paths. also, the path-cost functions have as argument the path-flow vectors h .

Link-cost functions have as natural argument the link-flow vector v , which in turn can be expressed through the path-flow vector; i.e, $v = \Delta \cdot h$.

For instance, for the O-D pair $w_1 = (6, 9)$:

$$\begin{aligned}
C_1^{w_1}(h) &= s_1(v_1) + s_4(v_4) = s_1(h_1^{w_1} + h_3^{w_1} + h_1^{w_2}) + s_4(h_1^{w_1} + h_4^{w_1} + h_2^{w_3}) \\
&= 50 + h_1^{w_1} + h_3^{w_1} + h_1^{w_2} + 10 \cdot (h_1^{w_1} + h_4^{w_1} + h_2^{w_3}) \\
&= 50 + 11 \cdot h_1^{w_1} + h_3^{w_1} + h_1^{w_2} + 10 \cdot h_4^{w_1} + 10 \cdot h_2^{w_3} \\
C_2^{w_1}(h) &= s_3(v_3) + s_2(v_2) = \dots \\
C_3^{w_1}(h) &= s_1(v_1) + s_6(v_6) + s_2(v_2) = \dots \\
C_4^{w_1}(h) &= s_3(v_3) + s_5(v_5) + s_4(v_4) = \dots \\
C_1^{w_2}(h) &= s_1(v_1) = \dots \\
C_2^{w_2}(h) &= s_3(v_3) + s_5(v_5) = \dots \\
C_1^{w_3}(h) &= s_2(v_2) = \dots \\
C_2^{w_3}(h) &= s_5(v_5) + s_4(v_4) = \dots
\end{aligned}$$

Given some feasible vector of path-flows h , the shortest cost route will be simply defined for each O-D pair as

$$u_{w_j}^*(h) = \min\{C_1^{w_j}(h), \dots, C_{n_j}^{w_j}(h)\}, \quad j = 1, 2, 3, \dots, |W|$$

where the set of route costs $\{C_1^{w_j}(h), \dots, C_{n_j}^{w_j}(h)\}$ (assumed to be in number n_j for O-D pair $w_j \in W$) is formed by calculating them on the whole acyclic set of routes for w_j .

Thus,

$$\begin{aligned}
u_{w_1}^*(h) &= \min\{C_1^{w_1}(h), C_2^{w_1}(h), C_3^{w_1}(h), C_4^{w_1}(h)\} \\
u_{w_2}^*(h) &= \min\{C_1^{w_2}(h), C_2^{w_2}(h)\} \\
u_{w_3}^*(h) &= \min\{C_1^{w_3}(h), C_2^{w_3}(h)\}
\end{aligned}$$

For instance, for

$$h' = (1, 1, 2, 2, 10, 0, 2, 3) \rightarrow v' = (13, 5, 3, 6, 5, 2)$$

$$v' = \Delta \cdot h' \Leftarrow \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \\ 2 \\ 10 \\ 0 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 13 \\ 5 \\ 3 \\ 6 \\ 5 \\ 2 \end{bmatrix}$$

$$\begin{aligned}
u_{w_1}^*(h') &= \min\{63 + 60, 30 + 55, 63 + 12 + 55, 30 + 15 + 60\} \\
&= 30 + 55 = 85, \quad \text{corresponding to path 2.} \\
u_{w_2}^*(h') &= \min\{63, 30 + 15\} = 45, \quad \text{for path 2.} \\
u_{w_3}^*(h') &= \min\{55, 60 + 15\} = 55, \quad \text{for path 1.}
\end{aligned}$$

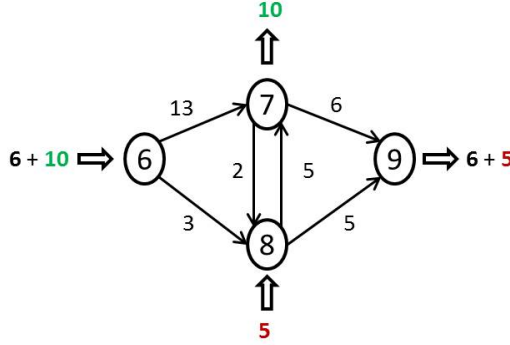


Figure 2: Link flows

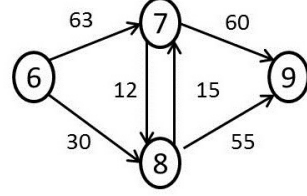


Figure 3: Link costs

5 Problems equivalent to formulations of equilibrium (Wardrop 1st principle)

5.1 Formulation 1. Complementarity problem.

$$w_1 : \left. \begin{aligned} C_1^{w_1}(h) - u_{w_1}^*(h) > 0 &\Rightarrow h_1^{w_1} = 0 \\ C_2^{w_1}(h) - u_{w_1}^*(h) > 0 &\Rightarrow h_2^{w_1} = 0 \\ C_3^{w_1}(h) - u_{w_1}^*(h) > 0 &\Rightarrow h_3^{w_1} = 0 \\ C_4^{w_1}(h) - u_{w_1}^*(h) > 0 &\Rightarrow h_4^{w_1} = 0 \end{aligned} \right\} \text{bad routes are not used} \quad (1)$$

$$\left. \begin{aligned} h_1^{w_1} > 0 &\Rightarrow C_1^{w_1}(h) - u_{w_1}^*(h) = 0 \\ h_2^{w_1} > 0 &\Rightarrow C_2^{w_1}(h) - u_{w_1}^*(h) = 0 \\ h_3^{w_1} > 0 &\Rightarrow C_3^{w_1}(h) - u_{w_1}^*(h) = 0 \\ h_4^{w_1} > 0 &\Rightarrow C_4^{w_1}(h) - u_{w_1}^*(h) = 0 \end{aligned} \right\} \text{used routes must be good} \quad (2)$$

$$w_2 : \left. \begin{aligned} C_1^{w_2}(h) - u_{w_2}^*(h) > 0 &\Rightarrow h_1^{w_2} = 0 \\ C_2^{w_2}(h) - u_{w_2}^*(h) > 0 &\Rightarrow h_2^{w_2} = 0 \end{aligned} \right\} \quad (3)$$

$$\left. \begin{aligned} h_1^{w_2} > 0 &\Rightarrow C_1^{w_2}(h) - u_{w_2}^*(h) = 0 \\ h_2^{w_2} > 0 &\Rightarrow C_2^{w_2}(h) - u_{w_2}^*(h) = 0 \end{aligned} \right\}$$

$$w_3 : \text{ Try to develop this case by yourself} \quad (4)$$

This formulation leads to a complementary condition:

$$h_r^{w_j} \cdot (C_r^{w_j}(h) - u_{w_j}^*(h)) = 0, \quad r \in \Gamma_{w_j}, \quad w_j \in W \quad (5)$$

Solving a traffic assignment problem is thus equivalent to:

FIND SOME $h \in \Omega$ so that CONDITIONS (1-4) ARE MET

or equivalently: CONDITIONS (5) ARE MET.

5.2 Formulation 2. Variational inequality problem.

If h (or v) verifies FORMULATION 1 (i.e., conditions (1-4)), then h (or v) verifies also the following condition (VI). The converse is also true.

- (VI) for path-flows and path-costs in the example net. Given any feasible path-flow vector $\hat{h}$:

$$\begin{aligned} 0 \leq & C_1^{w_1}(h) \cdot (\hat{h}_1^{w_1} - h_1^{w_1}) + \dots + C_4^{w_1}(h) \cdot (\hat{h}_4^{w_1} - h_4^{w_1}) + \\ & + C_1^{w_2}(h) \cdot (\hat{h}_1^{w_2} - h_1^{w_2}) + C_2^{w_2}(h) \cdot (\hat{h}_2^{w_2} - h_2^{w_2}) \\ & + C_1^{w_3}(h) \cdot (\hat{h}_1^{w_3} - h_1^{w_3}) + C_2^{w_3}(h) \cdot (\hat{h}_2^{w_3} - h_2^{w_3}) \end{aligned}$$

- (VI) for link-flows and link-costs in the example net. For any feasible link-flow vector $\hat{v} = \Delta \cdot \hat{h}$:

$$0 \leq s_1(v_1)(\hat{v}_1 - v_1) + s_2(v_2)(\hat{v}_2 - v_2) + \dots + s_6(v_6)(\hat{v}_6 - v_6)$$

5.3 Formulation 3. Optimization problem. Path-based

In many practical applications, link-cost functions $s_i(\cdot), i = 1, \dots, 6$ are: 1) non-negative, 2) non-decreasing, 3) continuous and have as only argument its own link flow, i.e.

$$\begin{aligned} s_i : \quad & \mathbb{R} \rightarrow \mathbb{R} \\ & x \rightarrow s_i(x) \quad i = 1, \dots, 6 \end{aligned}$$

then, PROBLEMS 1 and 2 (which are both equivalent each other) can be expressed as a non-linear optimization problem.

(A typical example for $s_i(\cdot)$ is the classical BPR function, a power function. Class slides show functional forms used in professional practice and will not be repeated here).

For the example:

$$\min_{v,h} \int_0^{v_1} s_1(x)dx + \int_0^{v_2} s_2(x)dx + \cdots + \int_0^{v_6} s_6dx$$

s.t.:

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} h_1^{w_1} \\ h_2^{w_1} \\ h_3^{w_1} \\ h_4^{w_1} \\ h_1^{w_2} \\ h_2^{w_2} \\ h_1^{w_3} \\ h_2^{w_3} \end{bmatrix}$$

$$h_1^{w_1} + h_2^{w_1} + h_3^{w_1} + h_4^{w_1} = 6$$

$$h_1^{w_2} + h_2^{w_2} = 10$$

$$h_1^{w_3} + h_2^{w_3} = 5$$

$$h_i^{w_j} \geq 0, \quad i = 1, \dots, 6, \quad j = 1, \dots, 3$$

where, of course,

$$\begin{aligned} \int_0^{v_1} s_1(x)dx + \int_0^{v_2} s_2(x)dx + \cdots + \int_0^{v_6} s_6dx = \\ = 50v_1 + \frac{v_1^2}{2} + 50v_2 + \frac{v_2^2}{2} + 5v_3^2 + 5v_4^2 + 10v_5 + \frac{v_5^2}{2} + 10v_6 + \frac{v_6^2}{2} \end{aligned}$$

The multicommodity O-D-based (or O-based) link-flow formulation can also be used because, in our case, $s_i(x) \geq 0, \forall i$.

5.4 Formulation 3'. Optimization problem. Origin-based.

$$\begin{aligned}
\min_v \quad & 50v_1 + \frac{v_1^2}{2} + 50v_2 + \frac{v_2^2}{2} + 5v_3^2 + 5v_4^2 + 10v_5 + \frac{v_5^2}{2} + 10v_6 + \frac{v_6^2}{2} \\
\text{s.t.:} \quad & v_1 = v_1^6 + v_1^8 \\
& v_2 = v_2^6 + v_2^8 \\
& \dots \\
& v_6 = v_6^6 + v_6^8
\end{aligned} \tag{6}$$

$$A \begin{bmatrix} v_1^6 \\ v_2^6 \\ \dots \\ v_6^6 \end{bmatrix} = \begin{bmatrix} 16 \\ -10 \\ 0 \\ -6 \end{bmatrix}, \quad v_i^6 \geq 0, i = 1, \dots, 6 \tag{7}$$

$$A \begin{bmatrix} v_1^8 \\ v_2^8 \\ \dots \\ v_6^8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \\ -5 \end{bmatrix}, \quad v_i^8 \geq 0, i = 1, \dots, 6$$

Total: 18 variables and 14 equality constraints.

Notice that constraints (8) are just definitional constraints. This means that each v_j could be replaced in the objective function by its expression in terms of v_j^6, v_j^8 and the total number of variables and constraints would be reduced.

5.5 Formulation 3''. Optimization problem. Destination-based.

Would you be able to rewrite the previous optimization problem in subsection 5.4 using the destination-based multicommodity formulation?

5.6 Formulation 3'''. Optimization problem. O-D-based.

Would you be able to rewrite the previous optimization problem in subsection 5.4 using the O-D-based multicommodity formulation?

Could you compare the O-D-based multicommodity formulation with the path-based formulation in subsection 5.3?

6 Solution of the worked example

Solved in AMPL. Model file:

```

set nusos;
set centr within nusos;
set links within ( nusos cross nusos );
set origens within centr;
set destins within centr;
set odpair within ( origens cross destins );
set destperorig { i in origens } :=
setof { (i1, j1) in odpair : i = i1 } j1;

param g{odpair} >0;
param t0{links};
param c{links};
param cap{links};
param Tdreta { i in nusos, k in origens }:=
if i in destperorig[k] then -1.0*g[k, i]
else
if i = k then sum {j in destperorig[k]} g[k, j] else 0;

node N {i in nusos, k in origens}: net_out = Tdreta[i, k];

arc v_k { (i, j) in links, k in origens } >= 0,
from N[i, k], to N[j, k] ;

var v { (i, j) in links };

subject to flux_total { (i, j) in links }:
v[i, j] = sum { k in origens } v_k[i, j, k];

minimize Vg: sum { (i, j) in links } v[i, j]*t0[i, j];
minimize Vn1: sum { (i, j) in links } (v[i, j]*c[i, j] + 0.5*cap[i,j]*v[i, j]^2);

```

Data file:

```

set nusos:= 6 7 8 9 ;
set centr:= 6 7 8 9 ;
set links:= (6,7) (6,8) (7,8) (7,9) (8,7) (8,9);

set origens:= 6 8;
set destins:= 7 9;
set odpair:= (6,9) (6,7) (8,9);
param g:=
6 9 6
6 7 10
8 9 5;

```

```
param c:=  
6 7 50  
6 8 0  
7 8 10  
7 9 0  
8 7 10  
8 9 50;  
param cap:=  
6 7 1  
6 8 1  
7 8 10  
7 9 10  
8 7 1  
8 9 1;
```

Solution:

Total link flows v_a^* and link travel times $s_a(v_a^*)$ at equilibrium:

```
MINOS 5.51: optimal solution found.
5 iterations, objective 818.4583333
Nonlin evals: obj = 8, grad = 7.
ampl: display v; # total link flows
v :=
6 7    0
6 8   16
7 8    0
7 9    3.41667 = 41/12
8 7   13.41667 = 161/12
8 9    7.58333 = 91/12
;

ampl: display { (i, j) in links } (c[i, j] + cap[i,j]*v[i, j]);
# link travel times
c[i,j] + cap[i,j]*v[i,j] :=
6 7    50
6 8    16
7 8    10
7 9   34.1667 = 205/6
8 7   23.4167 = 281/12
8 9   57.5833 = 691/12

ampl: display v_k; # flows per-origin at links
v_k :=
6 7 6    0
6 7 8    0
6 8 6   16
6 8 8    0
7 8 6    0
7 8 8    0
7 9 6    0
7 9 8    3.41667 = 41/12
8 7 6   10
8 7 8    3.41667 = 41/12
8 9 6    6
8 9 8    1.58333 = 19/12
```

We can now apply formulas in section 4 to calculate the path costs at equilibrium. Notice that:

$$C_1^{w_1}(h^*) = 50 + \frac{410}{12} = \frac{501}{6}, \quad C_2^{w_1}(h^*) = 16 + \frac{691}{12} = \frac{883}{12},$$

$$C_3^{w_1}(h^*) = 50 + 10 + \frac{691}{12} = \frac{1411}{12}, \quad C_4^{w_1}(h^*) = 16 + \frac{281}{12} + \frac{410}{12} = \frac{883}{12}$$

Paths 1 and 3 will carry no flow. O-D flow will split into paths 2 and 4 for O-D $w_1 = (6, 9)$

$$C_1^{w_2}(h^*) = 50, \quad C_2^{w_2}(h^*) = 16 + \frac{281}{12} = \frac{473}{12}$$

paths used: only path 2 for O-D $w_2 = (6, 7)$

$$C_1^{w_3}(h^*) = \frac{691}{12}, \quad C_2^{w_3}(h^*) = \frac{281}{12} + \frac{410}{12} = \frac{691}{12}$$

paths used: O-D flow will split into paths 1 and 2 for O-D $w_3 = (8, 9)$

Check that with this information and the flows per origin results, the path flow vector must be:

$$h^* = (0, 6, 0, 0, 0, 10, \frac{41}{12}, \frac{19}{12})$$

although in general it will not be possible to infer univocally equilibrium flows on paths from equilibrium flows on links. (i.e, several path flow vectors may correspond to a given link flow vector). See Sheffi, (1985) Urban Transportation Networks, (Prentice Hall), chapter 3, page 68, for an example.

7 Subproblem or All-or-Nothing problem at v^* for the worked example

At a flow vector v' , the All-or-Nothing problem (AoN in the following), or Subproblem, is formulated taking the link travel times evaluated at v' as link costs. Thus, if $v' = v^*$ is adopted as example, the resulting AoN problem at

v^* will be:

$$\begin{aligned}
& \min_y \quad 50y_1 + \frac{691}{12}y_2 + 16y_3 + \frac{410}{12}y_4 + \frac{281}{12}y_5 + 10y_6 \\
& \text{s.t.:} \\
& \quad y_1 = y_1^6 + y_1^8 \\
& \quad y_2 = y_2^6 + y_2^8 \\
& \quad \dots \\
& \quad y_6 = y_6^6 + y_6^8
\end{aligned} \tag{8}$$

$$A \begin{bmatrix} y_1^6 \\ y_2^6 \\ \dots \\ y_6^6 \end{bmatrix} = \begin{bmatrix} 16 \\ -10 \\ 0 \\ -6 \end{bmatrix}, \quad y_i^6 \geq 0, i = 1, \dots, 6 \tag{9}$$

$$A \begin{bmatrix} y_1^8 \\ y_2^8 \\ \dots \\ y_6^8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \\ -5 \end{bmatrix}, \quad y_i^8 \geq 0, i = 1, \dots, 6$$

which can be reformulated as two independent problems, one for each origin:

$$\begin{aligned}
& \min_{y^6} \quad 50y_1^6 + \frac{691}{12}y_2^6 + 16y_3^6 + \frac{410}{12}y_4^6 + \frac{281}{12}y_5^6 + 10y_6^6 \\
& \text{s.t.:} \\
& \quad y_1^6 = y_1^6 + y_1^8 \\
& \quad y_2^6 = y_2^6 + y_2^8 \\
& \quad \dots \\
& \quad y_6^6 = y_6^6 + y_6^8
\end{aligned} \tag{10}$$

$$\begin{aligned}
& \min_{y^8} \quad 50y_1^8 + \frac{691}{12}y_2^8 + 16y_3^8 + \frac{410}{12}y_4^8 + \frac{281}{12}y_5^8 + 10y_6^8 \\
& \text{s.t.:} \\
& \quad y_1^8 = y_1^6 + y_1^8 \\
& \quad y_2^8 = y_2^6 + y_2^8 \\
& \quad \dots \\
& \quad y_6^8 = y_6^6 + y_6^8
\end{aligned} \tag{11}$$

The first of the previous two problems is equivalent to calculate the shortest paths from 6 to 9 and from 6 to 7. The second, is equivalent to calculate the shortest path between 8 and 9.

Finding a solution of the subproblem (SUBPR) at v^* is necessary to evaluate the absolute gap at v^* .

The absolute gap for (SUBPR) at v^* will be

$$\sum_{i=1}^6 s_i(v_i^*)(y_i - v_i^*) = 50(y_1 - 0) + \frac{691}{12}(y_2 - \frac{91}{12}) + 16(y_3 - 16) + \frac{410}{12}(y_4 - \frac{41}{12}) + \frac{281}{12}(y_5 - \frac{161}{12}) + 10(y_6 - 0)$$

Since we are evaluating the absolute gap at v^* , we know beforehand that:

$$\sum_{i=1}^6 s_i(v_i^*)(y_i - v_i^*) = 0$$

at any solution y of the subproblem SUBPR.

Knowing that solutions of SUBPR6 can be found using the shortest paths for O-D pairs (6,9), (6,6), the following flow vectors y^6 can be solutions of SUBPR6:

- 1) using path 2 and 4 for O-D (6,9) and path 2 for path (6,7) : $y^6 = (0, 3, 16, 3, 13, 0)$
- 2) using path 2 for O-D (6,9) and path 2 for path (6,7) : $y^6 = (0, 6, 16, 0, 10, 0)$
- ...

Notice that solution 2) corresponds to an AoN loading whereas 1) does not.
Can you find more solutions for SUBPR6 ?
and for SUBPR8?